YUXI ZHANG

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RESEARCH PROFILE

PhD researcher in the Speech and Audio Processing Group at Imperial College London, supervised by Prof. Patrick A. Naylor. My research focuses on deep internal learning, test-time learning, and self-supervised signal enhancement: training neural networks directly from individual observations without external training data or pretraining. I developed an utterance-specific approach to speech dereverberation, studied acoustic mismatch, and evaluated downstream ASR on LibriSpeech. I am also exploring internal learning for speech bandwidth extension and ECG enhancement, with broader interests in data-efficient machine learning and robust speech and audio systems.

EDUCATION

Imperial College London — PhD in Electrical and Electronic Engineering
Oct 2023 – Oct 2027 (Expected) | Speech and Audio Processing Group | Supervisor: Prof. Patrick A. Naylor

Imperial College London — MSc in Communications and Signal Processing, Distinction
Oct 2022 – Oct 2023

University of Liverpool — BEng in Computer Science and Electronic Engineering, First Class Honours
Sep 2018 – Jun 2022 | Ranked 2nd in programme in Years 2–3

RESEARCH EXPERIENCE

Deep Internal Learning for Speech Dereverberation
PhD Research | Speech and Audio Processing Group, Imperial College London | Oct 2023 – Present

- Proposed a deep internal learning (DIL) approach to single-channel speech dereverberation, investigating whether a neural network can learn to reverse an unknown acoustic degradation from a single observation without clean references, external training data, or pretraining.
- Developed an utterance-specific learning framework that constructs an internal training task using an auxiliary acoustic degradation model, training a compact multi-scale CNN (~0.21M parameters) from scratch on each observed utterance.
- Investigated how the relationship between the training-specific and true room impulse responses influences dereverberation performance, identifying the impact of acoustic mismatch through a structured sensitivity analysis.
- Benchmarked DIL against classical unsupervised (WPE), pretrained discriminative (DeepFilterNet), and generative (BUDDy) methods across three speech corpora, demonstrating improvements under matched conditions and characterising performance limitations under acoustic mismatch.

Deep Internal Learning for Speech Bandwidth Extension
Ongoing PhD Research | Imperial College London | 2026 – Present

- Investigating whether frequency-dependent structure learned from a single speech utterance can be transferred from observed lower-frequency bands to reconstruct previously unseen high-frequency content, without external training data or pretraining.
- Developed CNN-based internal frequency extrapolation and hybrid spectral band replication (SBR) approaches; preliminary experiments demonstrate the potential of utterance-specific learning for high-frequency reconstruction, with improvements over a conventional SBR baseline on a small development set.

Deep Internal Learning for ECG Signal Enhancement
Ongoing PhD Research | Imperial College London | 2026 – Present

- Investigating the extension of deep internal learning (DIL) from speech processing to ECG enhancement, exploring self-supervised learning from individual noisy recordings without external training data or pretrained models.
- Explored two internally constructed learning strategies: degradation-based DIL, which learns to reverse additional synthetic noise, and repetition-based DIL, which exploits recurring cardiac structures through beat-aligned masked reconstruction using a 2D CNN.
- Developed and evaluated single-recording learning pipelines under controlled noise conditions, examining waveform reconstruction, R-peak detection, and the influence of heartbeat alignment on enhancement performance.

Mitigation of Gender Bias in Text-to-Image Generation Models

MSc Thesis | Imperial College London | Oct 2022 – Oct 2023

- Developed an in-processing debiasing approach for Stable Diffusion using sampling-based probability estimation to selectively augment under-represented gender prompts.
- Reduced the average gender bias ratio from 0.325 to 0.053 (83.7% relative reduction) across profession-, scenario-, and trait-based prompt groups.

PUBLICATIONS & MANUSCRIPTS

- Yuxi Zhang and Patrick A. Naylor. “Dereverberation Using Deep Internal Learning: Speech Enhancement and ASR Evaluation.” International Workshop on Acoustic Signal Enhancement (IWAENC), 2026.
- Yuxi Zhang, Emilie d’Olne, Vikas Tokala, and Patrick A. Naylor. “Deep Internal Learning for Single-Channel Speech Dereverberation.” European Signal Processing Conference (EUSIPCO), 2025.
- Yuxi Zhang and Patrick A. Naylor. “Deep Internal Learning for Speech Dereverberation.” IEEE Signal Processing Open — manuscript under major revision, 2026.
- Yuxi Zhang. “A Deep Learning-Based Tool for Face Mask Detection and Body Temperature Measurement.” International Conference on Signal Processing and Information Security (ICSPIS), 2022.
- Yuxi Zhang. “Realizing Blood Glucose Prediction by Convolutional Recurrent Neural Networks with Residual Blocks.” International Journal of Machine Learning and Computing, vol. 12, no. 6, 2022.

INDUSTRY EXPERIENCE

Omega Point Research — AI-Driven Investment Research & Technology

AI Research Associate (Part-time) | London, UK | Sep 2025 – Present

- Developing an in-house, multi-agent AI research platform supporting buy-side investment workflows, integrating financial data collection, market intelligence, company monitoring, and automated valuation.
- Built specialised research tools including a data centre infrastructure tracker, market and social-media monitoring systems, an LLM industry tracker, and an automated discounted cash flow (DCF) pipeline that transforms collected company information into valuation models.

TECHNICAL SKILLS

Programming & Frameworks: Python, PyTorch, TensorFlow, MATLAB, C++, SQL, Java.

Machine Learning: Deep internal learning, self-supervised learning, test-time adaptation, CNNs, Transformers, diffusion models.

Signal Processing: Speech enhancement, dereverberation, bandwidth extension, room acoustics, RIR modelling, time-frequency analysis, ECG processing.

Research & Computing: GPU training, Linux, HPC, experimental design, quantitative evaluation.

Languages: English (fluent), Mandarin Chinese (native).